Magnetic fields

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Interpret magnetic field patterns and compass direction.

SUCCESS CRITERIA - I can...

1. Label north and south poles.
2. Trace external field direction from north toward south.
3. Explain why field-line drawings are models.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “magnetic field” and “pole” to explain your model or result. Sentence starters are optional.

magnetic field: a field that influences magnets and moving charges

pole: a north or south region of a magnet

superposition: adding displacements or fields at the same place and time

01 Warm-up: bring back an earlier idea

A total current of 10 A splits between two identical parallel branches.
Find each branch current.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $B_{net,\,x} = B_{1,\,x} + B_{2,\,x}$; $B_{net,\,y} = B_{1,\,y} + B_{2,\,y}$.
Add field components at the same point. A compass follows the net field. Outside a bar magnet, arrows run north to south.

Worked reasoning: Outside a bar magnet, field arrows run from the north pole toward the south pole. A compass's north-seeking end follows the local field direction.

Your turn: Draw curved external field lines and two small compass arrows at different points.

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Label N and S on the bar magnet.
Sketch external field arrows and the direction a nearby compass would point.

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: 1 2 3

04 Apply the physics

Place two bar magnets on a table without letting them touch.
Predict the interaction when N faces N, then when N faces S.
Sketch both arrangements and draw the force on each magnet.

Sentence starter: I predict … . My model supports this because … .

Label poles; add force or field arrows.



Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A student draws field lines that cross at one point.
Explain why this is inconsistent with a single net field direction at that point.

Sentence starter: My claim is … . The diagram/data show …, so … .

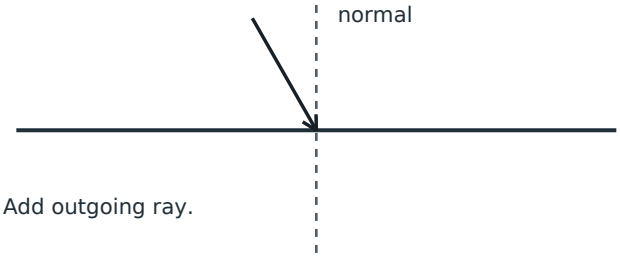
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A ray enters a medium with speed 1/4 of its original speed at an angle.
Sketch the bend and state whether frequency changes.
Label both media, the normal and ray directions.

Word help: frequency = number of cycles each second

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: 1 2 3

07 Exit check

What does closer spacing of field lines conventionally represent?

Sentence starter: I notice … . This supports … because … .



Marzano check - circle: 1 2 3

08 STRETCH: test the model

Cut a bar magnet into two pieces in a drawing.
Predict the poles on each piece and show the external field pattern qualitatively.

Sentence starter: I predict … . My model supports this because … .



Marzano check - circle: 1 2 3